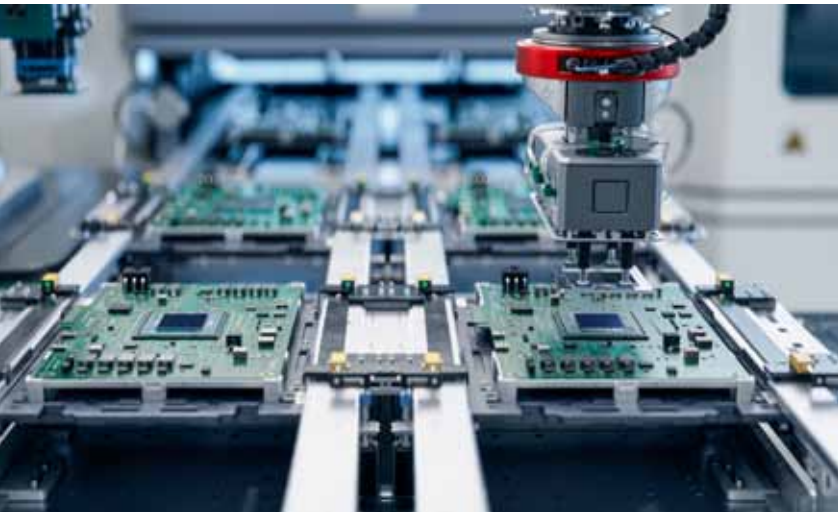


Harnessing AUTOMATED OPTICAL INSPECTION (AOI) TECHNOLOGIES For PCBA Manufacturing

Your smartphone detects defects smaller than a virus using AOI, laser, and X-ray systems. These machines don't just see—they think. These innovations, from AI-driven defect detection to Nano-CT systems revealing hidden flaws, reshape quality control in electronics manufacturing.



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Ensuring the quality and reliability of printed circuit board assemblies (PCBAs) is vital in electronics manufacturing. Automated optical inspection (AOI) systems are central to this effort, detecting defects throughout the assembly process. With the increasing complexity of PCB designs, AOI technologies have evolved, integrating optical, X-ray, and laser systems to improve detection. This article examines the strengths and applications of these three leading AOI technologies in PCBA production.

Optical AOI: The foundation of visual inspection

Optical AOI systems use high-resolution

cameras and advanced algorithms to inspect PCBAs for surface-level defects, such as:

Component misplacement. Ensuring components are correctly aligned and positioned

Solder joint issues. Detecting insufficient, excessive, or bridged solder

Missing components. Identifying areas where components are absent

Polarity errors. Verifying the correct orientation for polarised components

These systems rely on visible light, typically supplemented by LED lighting, to enhance contrast and highlight specific features. Modern optical AOI machines employ advanced imaging techniques, such as 3D imaging, to capture depth information and improve accuracy in detecting subtle defects.

However, optical AOI has its limitations. It excels at inspecting surface mount technology (SMT) components. Still, it faces challenges with solder joints hidden under packages like ball grid arrays (BGAs) or components on densely packed boards where line-of-sight is obstructed.

X-ray AOI: Peering beneath the surface

X-ray AOI systems address the limitations of optical inspection by enabling non-destructive analysis of solder joints and internal structures. They excel in identifying:

Hidden solder defects. Voids, cracks,